Lecture 15: Alternating Series and Conditional Convergence

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what is an alternating series.
- ☐ Apply the Alternating Series Test to determine if an alternating series converges.
- ☐ Apply the Alternating Series Estimation Theorem to estimate the sum of a convergent alternating series.
- ☐ Understand what is a conditionally convergent series.
- ☐ Understand how rearranging an infinite series affect its sum.

Alternating series

A series in which the terms are alternatively positive and negative is called an $\sum a_n$ alternating series.

Example:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

$$\sum_{n=1}^{\infty} \frac{(-1)^{n} 4}{2^n}$$

$$\sum_{n=1}^{\infty} (-1)^{n+1} n$$

Alternating Series Test

The Alternating Series Test

The series

$$\sum_{n=1}^{\infty}(-1)^{n+1}u_n = u_1 - u_2 + u_3 - u_4 + \cdots$$

converges if the following conditions are satisfied:

1. The u_n's are all positive.
2. The u_n's are eventually nonincreasing.
3. $u_n \to 0$.

Alternating Series Test

Alternating Series Test

Example: $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n}$ (Alternating harmonic series)

The Alternating Series Test

Example: $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{10n}{n^2+16}$

Error in Approximating Alternating Series

Suppose that a_k is a sequence satisfying

$$a_0 \geq a_1 \geq \cdots \geq a_n \geq a_{n+1} \geq \cdots \geq 0,$$

with $\lim_{n\to\infty} a_n = 0$.

Alternating Series Estimation Theorem

The Alternating Series Estimation Theorem

If the alternating series $\sum_{n=1}^{\infty}(-1)^{n+1}u_n$ satisfies the three conditions of the Alternating Test, then for $n \geq N$,

$$s_n = u_1 - u_2 + \cdots + (-1)^{n+1}u_n$$

approximates the sum L of the series with an error whose absolute value is less than u_{n+1}, the absolute value of the first unused term. Furthermore, the sum L lies between any two successive partial sum s_n and s_{n+1}, and the remainder, $L - s_n$, has the same sign as the first unused term.

Alternating Series Estimation Theorem

Example: $\sum_{n=0}^{\infty}(-1)^n \frac{1}{2^n}$

Conditional Convergence

A series that is convergence but not absolutely convergent is called conditional convergent.

Conditional Convergence

Example: For a positive constant p, consider the alternating p-series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^p}.$$

Rearranging Series

The Rearrangement Theorem for Absolutely Convergent Series

If $\sum_{n=1}^{\infty} a_n$ converges absolutely, and $b_1, b_2, \ldots, b_n, \ldots$ is any arrangement of the sequence $\{a_n\}$, then $\sum_{n=1}^{\infty} b_n$ converges absolutely and

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n.$$

So we can always rearrange the terms of an absolutely convergent infinite series. But can we do the same for a conditional convergent series?

Rearranging Series

Consider the alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

Summary

1. **The *n*th-Term Test for Divergence:** Unless $a_n \to 0$, the series diverges.
2. **Geometric series:** $\sum ar^n$ converges if $|r| < 1$; otherwise it diverges.
3. ***p*-series:** $\sum 1/n^p$ converges if $p > 1$; otherwise it diverges.
4. **Series with nonnegative terms:** Try the Integral Test or try comparing to a known series with the Direct Comparison Test or the Limit Comparison Test. Try the Ratio or Root Test.
5. **Series with some negative terms:** Does $\sum |a_n|$ converge by the Ratio or Root Test, or by another of the tests listed above? Remember, absolute convergence implies convergence.
6. **Alternating series:** $\sum a_n$ converges if the series satisfies the conditions of the Alternating Series Test.